

Practice problems #4

1. True or False. Justify. All of the following matrices are real $n \times n$ matrices.
 - (a) If A and B are similar, $\det A = \det B$.
 - (b) If $A = QR$ with Q invertible, the A is similar to RQ .
2. (This is #8 of Class #3 handout.) Diagonalize the following matrix, if possible.

$$\begin{bmatrix} 7 & 4 & 16 \\ 2 & 5 & 8 \\ -2 & -2 & -5 \end{bmatrix}$$

Hints or Answers on page 2.

#1(a): True. Justify!

#1(b): True. Note that _____ QR _____ = RQ .

#2: The following answer is not unique depending upon which eigenvector you use and which order you list the corresponding eigenvalues. An example of an answer might be:

$$P = \begin{bmatrix} 2 & 1 & 4 \\ 1 & -1 & 0 \\ -1 & 0 & -1 \end{bmatrix}, \quad D = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 3 \end{bmatrix}$$